Request for Proposal:

Additional Features for the Carleton LRT Station

Carleton University

ECOR3800B: Engineering Economics

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1 Introduction

This request for proposal is for the addition of numerous features for the Carleton LRT train station. The goal is to include budget, timeline, and projected requirements for the infrastructure for bicycles to park at the station, solar powered food trucks, local Ottawa artwork, parking lot renovations, and various disability accommodation features.

1.1 Team Description

Surya Nareshan will be the lead project manager to be in charge of the progress and interim reports over the course of the project, along with the creation of the AON and Gantt charts, ensuring every task was on schedule. Max Campbell will be the lead electrical engineer to provide consultation on any electrical considerations and to work with the contractors to choose the most efficient designs. Similarly, Hashan Singh Dhaliwal and Will Abbott will be the lead civil and mechanical engineers, to provide consultation to Surya and the contractors to assure every design constructed will be safe and efficient. Finally, Wassim Djari will be the financial advisor, researching and comparing many different options and will help to provide accurate and reasonable estimates for all the necessary costs. Along with these members, many different subcontractors services will be chosen for the construction jobs. For the solar panels, the company Polaron will provide all of the necessary tools, equipment, and labour [1]. Countryside Gazebos will be in charge of the construction of the pavilion [2]. Faulkner Electric Ltd. was chosen for electrical consultation and installation. [3]. For any other general contractor work, Ottawa General Contractors proved to be the most reliable choice [4].

2 Preliminary Assessment and Design of the Project.

The project was designed with the goal of having an eco-friendly location, which is both enjoyable to visit, no matter the weather, and easy to maintain. Each sub-project was approached with the objective of making the project appealing while aligning with the social, environmental, and ethical standards upheld by the university. The workflow path and various dependencies of the project are displayed in an AON diagram (refer to Figure 1 in Appendix B). Each activity's detailed work breakdown structure will be explained for a full understanding of the project.

2.1 Repaving of the parking lot

Carleton's parking lot 4, located directly beside the train line, is the desired location for the project. Currently, this location is in poor shape due to the large amount of construction and overall lack of maintenance. As such, it will require renovations in order to make the proposed project more feasible. It was decided to repave using asphalt due to it being more cost efficient and its resistance to cold weather [5]. There are no dependencies for this activity. This activity involves receiving the required permits, surface milling, installing the required underground infrastructure, paving, and installing the lighting. This work will be divided between a number of different contractors for the paving, electrical work, and installations. Overall, this sub-project should take an estimated 32 days. From research, asphalt needed to pave the parking lot costs around $4.00 per square foot, while labour costs can be estimated to also cost $4.00 per square foot [6]. The parking lot is 160 feet x 75 feet or 12 000 square feet. It was decided that 8 street lamps were to be installed to provide a good amount of visibility without being too harshly bright. Each pole requires 2 light bulbs which cost $150.00 and with each pole itself costing $500.00 [7][8]. The average cost of labour to install these lights was estimated to be approximately $1000.00 per pole [9]. Therefore, this portion of the project comes out to $110 800.00. All sample calculations for cost can be found in Appendix A.

2.2 Pavilion

In order to enhance the experience and usage of the new area, a seating area must be provided. This area should ideally be covered as well in order to encourage year round use. In this regard, a 24' x 24' wooden pavilion is being proposed as the centerpiece for the revamped parking lot. This structure was specifically chosen, as it is made out of a renewable resource, wood, which aligns with the environmental goals of the project as opposed to alternative options like vinyl pavilions. The cost of this pavilion is estimated to come out to $26 600.00. There will be lighting and electric space heaters in the pavilion to improve the experience during various times and seasons, all while being connected to the mounted solar panels [10]. Each heater will cost $185.00 and the labor to build them will cost $130.00 per hour from a general electrician [11]. The wiring of each component is assumed to cost $500.00 for 2000 feet [12]. This project will be broken into various stages with some dependencies on other sub-projects. The site assessment and design of the pavilion, as well as the acquiring of the materials are free of dependencies. The installation, however, is dependent on the completion of the paving as the pavilion installation will need the

pavement to be complete to build. After installing the pavilion, the electrical work must be completed which is dependent on the installation of the solar panels in order to properly connect the two systems. Finally, finishing touches such as staining are dependent on the completion of the electrical installations. Overall, this sub-project timeline is approximately 36 days without the dependencies [12]. The total cost of this sub-project comes to $76 100.00 with estimations.

2.3 Solar panel

A major portion of the project is implementing support for food trucks while minimizing the university's carbon footprint. This can prove to be a difficult task as a lot of energy is required to run food trucks, requiring up to 10 kW of energy per truck [13]. On top of the energy requirement of the food trucks, energy is required for the lighting around the parking lot and in the pavilion, and the space heaters being installed. The project is estimated to require 58 kW of energy to be supplied. In order to cleanly offset these energy demands, a solar panel system will be installed on the roof of the pavilion and around the lot. This would cost approximately $22 500 for a 7.5 kW system, and thus $180 000 for the eight required systems to completely offset the energy demand (as seen in Table 1) [1]. This job will be outsourced to a sub-contractor, Polaron, where the project payment will include the material and labour costs, and thus need not be accounted for in this project's budget. There are a number of steps required for the installation process including a site assessment, design process, applications to the various permits and programs for green energy, procurement of the system, installation, and the finalization of the system, which include monitoring of the panels, and finishing any applications and permits needed. The only dependency in the project is the installation, which requires the installation of the pavilion to be complete. Overall this sub-project takes 105 days [1].

2.4 Accessibility features to train station

Another consideration for this project is accessibility. It is important that any person who may experience some difficulties with or impairment of mobility can still easily, safely, and comfortably access the station. Therefore, a large wheelchair ramp was chosen to be placed at the entrance of the station. A ramp length of 20 feet was chosen since the train doors are elevated off the ground. The ramp will be made of concrete due to its strong and durable properties. Concrete also can last for years without needing substantial repairs. It is estimated that a ramp of this size and material would cost about $250 per linear foot [14]. Two days are allotted for the engineers

and contractors to assess the site and draft a design of the ramp. The amount of concrete needed for the ramp should be ordered 5 days in advance to allow for the company time to prepare [15]. It was decided that Tomlinson was the most reputable company to order concrete from [16]. Once the concrete has arrived, the construction of the ramp is estimated to take approximately 2 work days to complete [17]. The labour cost over these days is generally $90 an hour [14]. None of these activities are dependent on other sub-projects.

Another proposed disability feature is automatic doors. Two sets of double doors were deemed appropriate. This was chosen largely in order to protect against the harsh climate. The space in between the two sets of doors provides insulation for the station, keeping it cool in the summer months and warmer in the winter months. Each door should cost approximately $1000.00 [18]. The parts were given a week buffer to arrive and the installation was assumed to be able to be completed in 2 days. This installation cost was assumed to also be $1000.00 per door, yielding a total of $14 440.00 [19].

2.5 Artwork

To consider the second most requested additional feature for the LRT, the goal of the project is to include a public art piece that would highlight the work of a local Ottawa artist. The proposal is to have a mural-like painting along the walls that go underground through the rails; not only showcasing Ottawa's artistic prowess and rich culture, but will also be thematically consistent with Carleton's famed tunnel art. This sub-project will be the last feature of the project to be implemented since its only dependency is the installation stage of the artwork. Installing the artwork during construction could cause issues or damage the artwork, therefore, it should be the last to be implemented. The first stage of this aspect of the project will be artist selection. With a plethora of artists to choose from and the fact that this sub-project is not dependent on any other sub-project, allowing for a relatively significant slack time. The earliest finish will be arbitrarily chosen to be 20 days to allow for community input and time to consider all options fully. The next step would be hiring and providing a contract to the chosen artist and providing them with time to create a design. 10 days was estimated to provide artists with sufficient time to come up with something creative and representative of the local community. The next activity for the artwork sub-project would be procurement. Generally, when hiring artists to paint murals, the paint equipment is bought for the job [20]. Necessary equipment include: paint tray, paintbrushes,

painter's tape, sponge, bucket, drop cloths and necessary materials include: acrylic artist paint, primer, and varnish [21]. The final activity for the subproject would be the artist's responsibility to finish the mural. The allocated time for their contract will be 15 days.

Cost-wise, the equipment can be all relatively cheaply at Home Depot. An average 2 L paint tray is $3.50, painter's tape is about $10.00. Paint brushes vary in prices, so a budget of $50.00 would provide artists with some options. Drop cloths, buckets and sponges all go for relatively cheap would come to close $50.00 for 4 of each. [22]. Primer is about $83.39 a gallon [23], acrylic paint is estimated to be $20.00-100.00 per gallon [24], and varnish is about $60.00 a gallon [25]. Labour costs vary significantly per artist and can be arbitrary. On average, $40.00 per square feet is a typical cost with a baseline fee of around $300.00 [26]. It was assumed that the artist will have 2, 10 feet x 10 feet, or 200 square feet of room to paint. The total cost comes to $8 656.89.

2.6 Bike Racks

Without proper infrastructure, cyclists can often be seen parking in unwanted spaces, chaining their bikes to trees, fences, or railings. This can cause damage or obstruction to pedestrian or road traffic, as well as contribute to bike theft. Installing bike racks can help to reduce these activities and encourage eco-friendly transportation alternatives. Considering the space available near the LRT stop and the number of people who use bicycles for their daily commute, 3 cycle shelters with 42 racks, which could provide storage for up to 210 bikes, was chosen. R-8239-SS Stainless steel wave bike rack can secure up to 5 bikes with a U-lock between the frame and one wheel [27]. These bike racks cost $441.00 per unit. Stainless steel is resistant to corrosion and wear, therefore, it will remain functional over a longer service life [28]. Turin cycle shelter (SH1.CS4) is an open sided cycle shelter with a canopy style roof providing weatherproof storage [29]. It can accommodate up to 14 bike racks. All steel is standardly hot dip galvanized to prevent rust. Each shelter costs around $30,000. Since these items last over a long time, there are very little or no maintenance costs. The approximate area required for this facility is 1300 sq. ft, and there is an additional cost of $10 per sq. ft to pave this area. It would require 6-7 days for this area to be concrete paved which is dependent on the completion of the underground infrastructure installation. Delivery of racks and shelter would take around 4 weeks. It would take a maximum 1 week to install the racks and shelters, none of which is dependent on other activities. The total

time required for this is 5-6 weeks. The labour cost for all of this is estimated to be around 18.50 an hour, for an estimation of $160.00. The total cost is $124 522.00.

2.7 General Features

It was decided that a garden area could be included to add pleasant scenery for people waiting at the station. 30 flower beds will be planted around the station with the flowers costing approximately $100.00 and the flower bed costing $900.00 [30]. It is assumed that 10 labourers would be in charge of creating the flower bed themselves, costing an average of $18.50 an hour [31]. Two gardeners in charge of this task are estimated to cost around $19.20 an hour and are estimated to work 24 hours each [31]. Along with this, 25 picnic tables will be spread out around the paved over parking lot with each picnic table costing about $500.00. All together, this portion costs $16 421.60.

3 Critical Path Management Schedule

Activities will begin April 9th. The Gantt chart was created to depict expected timing and sequence of project activities following the aforementioned CPM schedule. Timing and schedule of activities were entirely based on the assumptions made in the previous section which can be summarized in the AON in Figure 1.

Project meetings should be held weekly to summarize the progress of each activity as defined in Section 2 with simple progress reports. Bi-monthly interim reports will be made by the manager of each team which should showcase the progress of activity, any issues encountered, and performance reviews to show to sponsors.

The major milestones will be the completion of each additional feature for the station and/or logistical aspects for the features. Refer to Gantt chart in Figure 2 and Table 2 which summarized the specific timelines and milestones for each activity.

The critical path is the one which follows the installation of the solar panels, expecting to have an earliest finish of 105 work days.

4 Professional, Social, Environmental, and Ethical Considerations

This project is for the public interest and will have a significant cost, thus, it must follow rigorous professional, social, and environmental considerations so that the station created is an inclusive and welcoming space. The professionalism is the first essential component of the project, making sure to adhere to expected high quality standards. The typical project management lifecycle should be meticulously followed and logical and organized work breakdown structure will be established. As mentioned in section 4, project milestones, project meetings, interim reports, and progress reports required like any other work environment.

Socially, it is paramount that a diverse and inclusive environment is created so that the station can be a safe space for all regardless of ethnicity, religion, disability, or any other factors. These factors were considered when creating this request for proposal. Features like the ramp and automatic doors are planned to allow people with wheelchairs to easily access the station. Another important social consideration is food restrictions with food trucks. Choosing food trucks that take into account the diversity of dietary restrictions, religious/ethnic considerations. For instance, making sure to provide halal and kosher options.

Building any form of infrastructure comes with environmental costs. For instance, the presence of food trucks and the need for power will inevitably contribute to climate change. To mitigate environmental impacts, the goal was to reduce the carbon footprint of food trucks. The solution is to develop electric hookups for the 4 food trucks and seating area. One of the primary goals of the project is to reduce the need for cars which contribute 11% of Canada's greenhouse emissions with the use of electric based trains [32]. The inclusion of infrastructure for bicycles to park is meant to incentivise the use of bicycles which are the most environmentally friendly option.

Finally, ethical considerations are fundamental for accountability and integrity. Another important ethical consideration is adhering to proper labour laws and treating employees, contractors, with proper respect and that all parties are properly compensated and work reasonable hours. It will be important to follow all government regulations when it comes to building infrastructure from having external inspections, following safety guidelines and protocol, following building codes and standards, etc. Engineering ethics must be taken very seriously for moral and legal grounds.

5 Preliminary Consideration of Major Sources of Uncertainty

When developing timelines for different activities of the project, there is a high degree of uncertainty. It is important for preliminary considerations for any major sources of uncertainty and to examine approaches to potentially address and mitigate those risks.

The main uncertainty is due to the geographical nature of the station; Ottawa has extreme weather which can greatly impact construction and other logistical components of the project. The first preventative measure was starting the project in early April to avoid the winter. The 10 year-average from 2010-2019 snowfall was 6 cm in April in Ottawa [33]. This is small but not negligible so winter considerations for construction must still be taken into account. Most uncertainties for the functionality of features after the project is complete must also consider the harsh winter weather. Preventative measures must be taken for all features. For example, the pavilion was designed to be covered to encourage year-round use. The other uncertainty is around the timelines since most provided start/finish dates were based on estimations from research.

6 Conclusion

The primary goal of this project is to redesign the Carleton LRT station to meet the community's desires so that it is an inclusive public space. The report scrutinizes the feasibility of each proposed feature. This proposal includes the breakdown of the components for each activity, estimates of costs for budgeting, and estimated timelines and dependencies for each respective activity. The proposed budget is $575 503.60 with the project expected to take 105 work days with an earliest estimated finish date of September 15th, 2023.

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Appendix A: Cost Calculations

A.1 Paving cost calculation

($4.00 per square foot [materials])(12 000 square feet)

+ ($4.00 per square foot [labour])(12 000 square feet)

+ ($150.00 per light)(2 lights per pole)(8 poles) + ($500.00 per pole)(8 poles) + ($1000

= $48 000.00 + $48 000.000 + $2 800.00 + $2 800.00 + $4 000.00

= $110 800

A.2 Pavilion cost calculation

($185.00 per heater)(12 heaters) + ($130.00 per hour)(15)(8 hour work days)(3 electricians)

+ ($500.00 per 2000 ft wire)(2000 ft wire)+ $26 600.00

= $2 220.00 + $46 800.00 + $500.00 + 26 600.00 = $76 100.00

A.3 Artwork cost calculation

$3.50 + $10.00 + $50.00 + $50.00 + $83.39 + $100.00 + $60.00 + $300.00 +

($40.00 per square foot)(200 square feet)

= $8 656.89

A.4 General cost calculation

(30 Flower beds)($100.00 for the flowers) + (30 Flower beds)($900.00 per bed)

+ (25 tables)($500.00 per table) + (2 gardeners)(19.20 per hour)(24 hours)

= $3 000.00 + $27 000.00 + $12 500 + $921.60 = $16 421.60.

A.5 Bike racks cost calculation

(42 bike racks)($441.00 per unit) + (3 shelters)($30,000.00 per unit)

+ ($10.00 per sq ft)(1300 sq ft) + ($18.50 an hour)(160 hours)

= $18 522.00 + $90 000.00 + $13 000.00 + $3 000.00

= $124 522.00

A.6 Accessibility features cost calculation

(4 doors)($1000.00 per door [materials]) + (4 doors)($1000 per door [labour])

+ ($250.00 per linear foot)(20 feet) + (16 hours)($90.00 per hour)

= $4 000.00 + $4 000.00 + $5 000.00 +$ 1 440.00 = $14 440.00

Table 1: Price breakdown including every activity of the project and its cost.

Project	Material Cost				Labour Cost				Total Cost	
	Break Down			Total	Break Down			Total	Total	Breakdown
Paving	Ashphault: $4 per sq ft	$48,000	Cost of Ashpault	$48,000.00	$4 per sq ft	$48,000	Labour	$48,000	$96,000.00	$48,000.00 Ashpault $48,000.00 Labour
Electrical	2 lights*8poles*150 dollars	$2,800.00	Cost of lighting	$9,520.00	$1000/pole	$8,000.00	light pole instillation	$54,800.00	$64,320.00	
	8 poles*500 dollars	$4,000.00	Cost of light poles							
	12 heaters*185 dollars	$2,220.00	Cost of heaters		$130/hour	$46,800.00	general eletrician (3 people)			
	$500/2000 ft	$500.00	Cost of Wire							
Pavillion	N/A				N/A				$26,600.00	Project cost, paid to hired company assume wages included
Solar Panel	N/A				N/A				$180,000.00	22500/unit*8 units Project cost, paid to hired company assume wages included
Automatic doors	1000/door*4 doors	$4,000.00	door opener	$4,000.00		$4,000.00	door instilation	$4,000.00	$8,000.00	
Ramps	$250/linear foot*20 feet	$5,000.00	cost of concrete	$5,000.00	0/hour*16 hou	$1,440.00	ramp insillation	$1,440.00	$6,440.00	
Artwork		$200.00	art materials	$200.00	$4/in^2		Artist fee			
Picnic Table/flower Bed	$500/table*25 table	$12,500.00	picnic tables	$39,500.00	$18.5/hour	$22,200.00	Labour (10 people)	$22,200.00	$61,700.00	
	$900/Flower Bed	$27,000.00	Flower Bed							
Gardener	$100/flower bed*30 flower beds	$3,000.00	Plants	$3,000.00	$19.20/hour	$921.60	Gardeners (2 workers)	$921.60	$3,921.60	
Bike Rack	$441/unit*42	$18,522.00	Bike Racks	$108,522.00	$10/sqr ft*130	$13,000.00	Paving	$16,000.00	$124,522.00	
	$30000/shelter*3	$90,000.00	Shelters		$18.50/hour	$3,000.00	general Labourers			
								Total Cost	$571,503.60	

Appendix B: AON Diagram

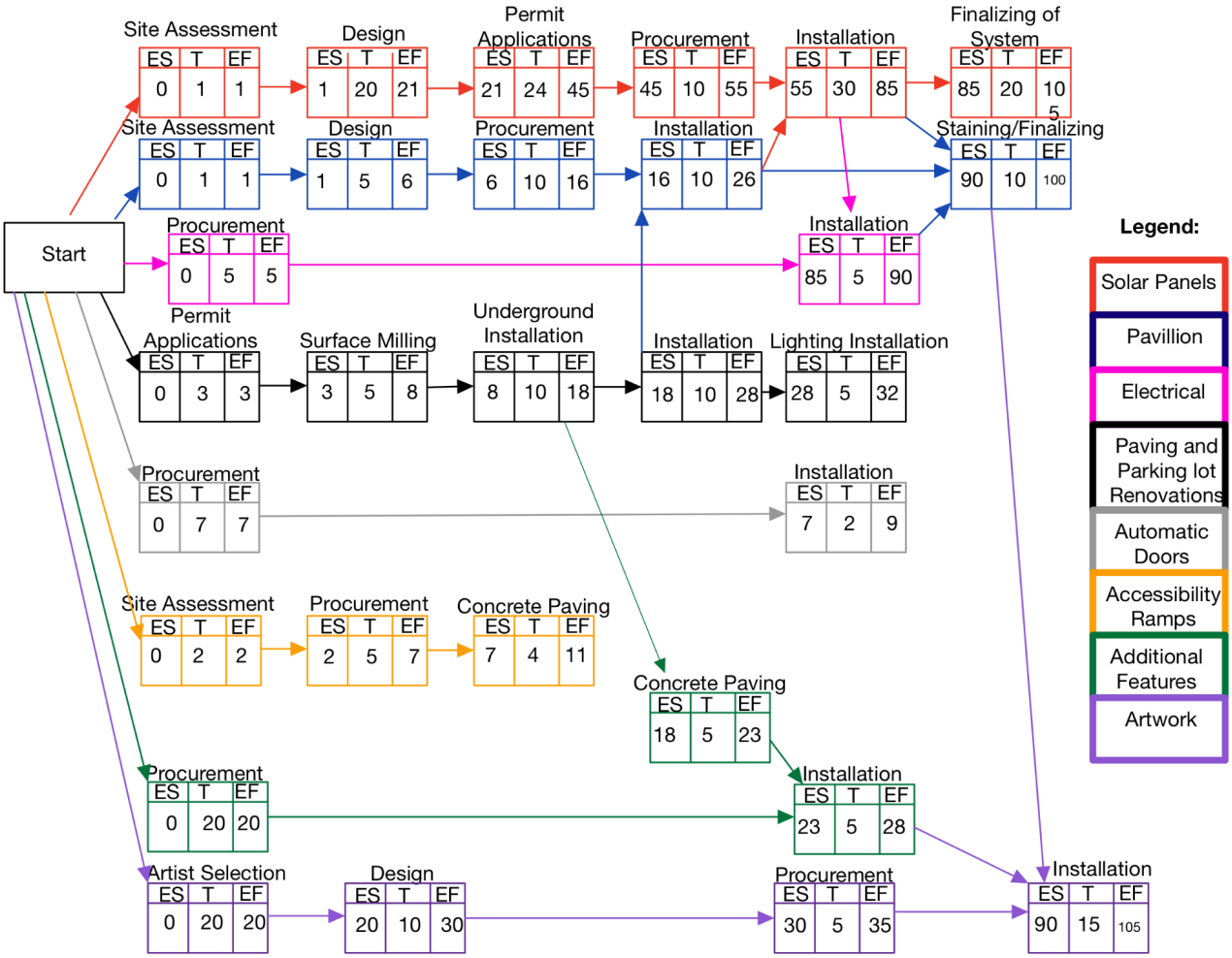

Figure 1: AON Diagram displaying all the activities and dependencies of this project [Campbell].

Appendix C: Gantt Chart

Figure 2: Gantt Chart for lifecycle of project and expected timelines for each activity based on the AON chart formulated from Figure 1.

Table 2: Summary of start and end dates of each activity as seen in the Gantt chart in Figure 2.

Name	Start Date	End Date	Duration
Acquiring of Picnic Tables	Aug 28, 2023	Aug 29, 2023	2 days
Artist Selection	Jul 10, 2023	Aug 04, 2023	20 days
Artwork Material Acquisition	Aug 21, 2023	Aug 25, 2023	5 days
Automatic Door Installation	Apr 19, 2023	Apr 20, 2023	2 days
Automatic Door Material Acquisition	Apr 10, 2023	Apr 18, 2023	7 days
Bike Shelter and Rack Installation	Aug 21, 2023	Aug 25, 2023	5 days
Bike Shelter Concrete Installation	Jun 05, 2023	Jun 09, 2023	5 days
Bike Shelter Order	Jul 24, 2023	Aug 18, 2023	20 days
Design of Artwork	Aug 07, 2023	Aug 18, 2023	10 days
Electrical Material Acquisition	May 01, 2023	May 05, 2023	5 days
Finalization of the System	Aug 07, 2023	Sep 01, 2023	20 days
Finalizing and Staining Pavilion	Aug 14, 2023	Aug 25, 2023	10 days
Flower bed Installation	Aug 28, 2023	Aug 29, 2023	2 days
Flower Bed Material Acquisition	Aug 24, 2023	Aug 25, 2023	2 days
Gardener Planting in Bed	Aug 30, 2023	Sep 01, 2023	3 days
Painting/Instillation of Artwork	Aug 28, 2023	Sep 15, 2023	15 days
Parking Lot Contractor and Permits	Apr 24, 2023	Apr 26, 2023	3 days
Parking Lot Lighting Installation	Jun 15, 2023	Jun 21, 2023	5 days
Pavilion Material Acquisition	May 23, 2023	Jun 05, 2023	10 days
Pavilion Design	May 16, 2023	May 22, 2023	5 days
Pavillion Electrical Installation	Aug 07, 2023	Aug 11, 2023	5 days
Pavilion Installation	Jun 06, 2023	Jun 19, 2023	10 days
Pavillion Site Assessment	May 15, 2023	May 15, 2023	1 day
Paving/Installation	May 22, 2023	Jun 02, 2023	10 days
Permits and Applications	May 09, 2023	Jun 09, 2023	24 days
Picnic Table Installation	Aug 30, 2023	Aug 31, 2023	2 days
Ramp Construction	Apr 19, 2023	Apr 24, 2023	4 days
Ramp Material Acquisition	Apr 12, 2023	Apr 18, 2023	5 days
Ramp Site Assessment and Design	Apr 10, 2023	Apr 11, 2023	2 days
Solar Panel Installation	Jun 26, 2023	Aug 04, 2023	30 days
Solar Panel Material Acquisition	Jun 12, 2023	Jun 23, 2023	10 days
Solar Panel Site Assessment	Apr 10, 2023	Apr 10, 2023	1 day
Solar Panel System Design	Apr 11, 2023	May 08, 2023	20 days
Staining/Final Touches Picnic Table	Sep 01, 2023	Sep 01, 2023	1 day
Surface Milling	May 01, 2023	May 05, 2023	5 days
Underground Electrical Installation	May 08, 2023	May 19, 2023	10 days